

| Research Question                                                                          | Data / Situation                                        | Statistical Test              | $H_0$                       | $H_1$                          | Test Statistic | Formula                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------|-----------------------------|--------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is a population mean different from a specified value when population $\sigma$ is known?   | One population, quantitative data, $\sigma$ known       | One-Sample Z-Test for Mean    | $\mu = \mu_0$               | $\mu \neq \mu_0$               | $Z$            | $\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$                                                                                                                                                                                                                                                                                                                           |
| Is a population mean different from a specified value when population $\sigma$ is unknown? | One population, quantitative data, $\sigma$ unknown     | One-Sample t-Test             | $\mu = \mu_0$               | $\mu \neq \mu_0$               | $t$            | $\frac{\bar{X} - \mu_0}{s/\sqrt{n}}$                                                                                                                                                                                                                                                                                                                                |
| Is a population proportion different from a specified value?                               | One population, binary/categorical outcome              | One-Proportion Z-Test         | $p = p_0$                   | $p \neq p_0$                   | $Z$            | $\frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$                                                                                                                                                                                                                                                                                                                       |
| Is a population variance different from a specified value?                                 | One population, quantitative data                       | Chi-Square Test for Variance  | $\sigma^2 = \sigma_0^2$     | $\sigma^2 \neq \sigma_0^2$     | $\chi^2$       | $\frac{(n - 1)s^2}{\sigma_0^2}$                                                                                                                                                                                                                                                                                                                                     |
| Are the means of two independent populations different?                                    | Two independent groups, quantitative outcome            | Independent Two-Sample t-Test | $\mu_1 - \mu_2 = 0$         | $\mu_1 - \mu_2 \neq 0$         | $t$            | <p>First test equality of variances using F-test <math>F = \frac{s_1^2}{s_2^2}</math></p> <p><i>if equal variances</i></p> $t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ <p><i>if unequal variances</i></p> $t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ |
| Are the variances of two populations different?                                            | Two populations, quantitative data                      | F-Test for Two Variances      | $\sigma_1^2/\sigma_2^2 = 1$ | $\sigma_1^2/\sigma_2^2 \neq 1$ | $F$            | $\frac{s_1^2}{s_2^2}$                                                                                                                                                                                                                                                                                                                                               |
| Are the proportions of two populations different?                                          | Two independent populations, binary/categorical outcome | Two-Proportion Z-Test         | $p_1 - p_2 = 0$             | $p_1 - p_2 \neq 0$             | $Z$            | $\frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$                                                                                                                                                                                                                                                               |

|                                                                |                                                             |                                 |                                   |                                           |          |                                           |
|----------------------------------------------------------------|-------------------------------------------------------------|---------------------------------|-----------------------------------|-------------------------------------------|----------|-------------------------------------------|
| Are the means of 3 or more independent populations different?  | 3+ independent groups, quantitative outcome                 | One-Way ANOVA                   | $\mu_1 = \mu_2 = \dots = \mu_k$   | At least one population mean is different | $F$      | $\frac{MS_{treatment}}{MS_{error}}$       |
| Are the means of 3 or more related treatments different?       | Same blocks measured under all treatments                   | Randomized Block ANOVA          | $\mu_1 = \mu_2 = \dots = \mu_k$   | At least one population mean is different | $F$      | $\frac{MS_{treatment}}{MS_{error}}$       |
| Does one categorical variable follow a specified distribution? | One categorical variable; observed vs. expected frequencies | Chi-Square Goodness-of-Fit Test | $p_i = p_{i0}$ for all categories | At least one $p_i \neq p_{i0}$            | $\chi^2$ | $\sum \frac{(O_i - E_i)^2}{E_i}$          |
| Are two categorical variables associated?                      | Two categorical variables in a contingency table            | Chi-Square Test of Independence | The two variables are independent | The two variables are associated          | $\chi^2$ | $\sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$ |